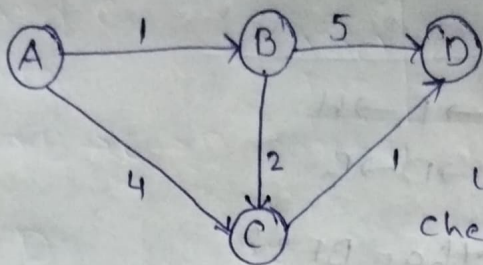


Lab-3: Uniform Cost Search



Here, edges have costs.

We have to find cheapest path from $A \rightarrow D$

Possible paths:

Path-1: $A \rightarrow B \rightarrow D$

$$\text{cost} = 1 + 5 = 6$$

Path-2: $A \rightarrow C \rightarrow D$

$$\text{cost} = 4 + 1 = 5$$

Path-3: $A \rightarrow B \rightarrow C \rightarrow D$

$$\text{cost} = 1 + 2 + 1 = 4$$

The cheapest path.

This has only 2 edges

has 3 edges

- So, more hops $\neq$ more cost.

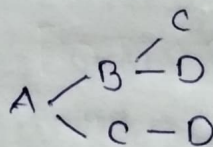
In BFS, $\rightarrow$ checks level-wise,

Level-1 $\rightarrow A$

Level-2 $\rightarrow B, C$

Level-3 $\rightarrow D$

it checks first



First goes as, $A \rightarrow B \rightarrow D$

cost = $1 + 5 = 6$ (more not the cheapest path).

So, uniform cost search

explores by

Smallest accumulated cost first.

1. graph = { 'A': [('B': 1), ('C': 4)],
 'B': [('D': 5), ('C': 2)],
 'C': [('D': 1)],
 'D': []
 }

Each neighbour has it's (neighbour, cost)

2. Declaring Queue,

queue = [(0, [start])]

↓ ↓
Total Path
cost

Initially → [(0, ['A'])]

↳ starting node

3. We continue below process until the Queue is empty.

Everytime, we sort the Queue.

queue.sort() → helps to keep cheapest cost path on top.

4. cost, path = queue.pop(0)

queue.top() → becomes the shortest cost one we pop that cheapest cost path.

POP → (0, 'A') | path = A

5. if path[-1] == goal

return cost, path

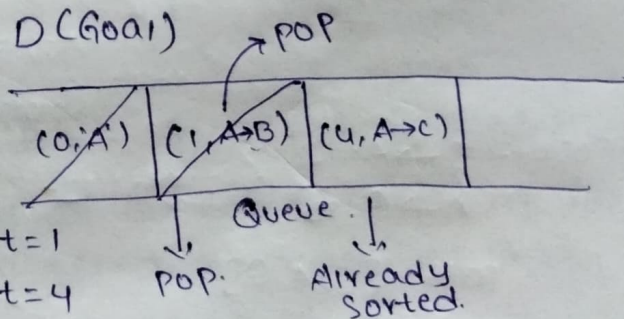
↓
We check if that goal is found, if yes, return cost & path.

path = A ≠ D (Goal)

Expand A
New Paths,

A → B cost = 1

A → C cost = 4



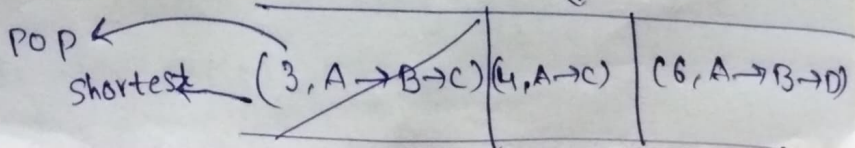
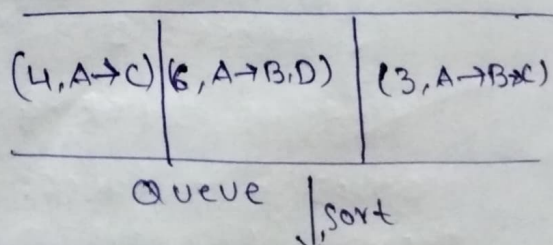
(1, A → B) → Not a goal.

Expand B

New paths,

A → B → D cost = 6

A → B → C cost = 3.



$A \rightarrow B \rightarrow C \neq D$ (Goal)

Expand C.

New path,

$A \rightarrow B \rightarrow C \rightarrow D$ cost = 4

$(4, A \rightarrow C)$	$(6, A \rightarrow B \rightarrow D)$	$(4, A \rightarrow B \rightarrow C \rightarrow D)$
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Queue

sort

$A \rightarrow C \neq D$

Expand C.

New path,

$A \rightarrow C \rightarrow D$ cost = 5

$(4, A \rightarrow C)$	$(4, A \rightarrow B \rightarrow C \rightarrow D)$	$(6, A \rightarrow B \rightarrow D)$
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POP

Smallest

$A \rightarrow B \rightarrow C \rightarrow D = D$ (Goal)

$(4, A \rightarrow B \rightarrow C \rightarrow D)$	$(6, A \rightarrow B \rightarrow D)$	$(5, A \rightarrow C \rightarrow D)$
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Goal found.

sort

$\therefore \text{Path} = A \rightarrow B \rightarrow C \rightarrow D$

$(4, A \rightarrow B \rightarrow C \rightarrow D)$	$(5, A \rightarrow C \rightarrow D)$	$(6, A \rightarrow B \rightarrow D)$
--	--------------------------------------	--------------------------------------

POP

Smallest